

Day-to-Day Temperature Changes and Internal Migration in the United States

Hyunsuk Oh*

September 29, 2026

Abstract

Research on climate as a local amenity has focused on the level and dispersion of temperatures, but the order in which daily temperatures arrive may also affect where people choose to live. This paper asks whether consecutive-day anomaly changes in temperature anomalies, holding their dispersion fixed, are associated with county-level internal migration in the United States. I measure the mean absolute change between consecutive daily anomalies, so that the remaining variation concentrates on how quickly anomalies reverse. Linking NOAA GHCN-Daily station records to IRS migration flows for 859 counties over eight filing-year pairs from 2011 to 2019, I relate migration to the three-year average of absolute day-to-day anomaly changes, controlling for within-month dispersion, the distribution of daily temperatures, precipitation, and seasonal amplitude, and including county and state-by-year fixed effects. A 1°C increase in consecutive-day anomaly changes is associated with a 0.39 percentage point rise in the out-migration rate, though the 95% confidence interval includes zero. The results show that the temporal ordering of temperatures can be distinguished empirically from dispersion, while evidence of a stable migration response remains limited.

Keywords: temperature variability; internal migration; climate amenities; adjustment costs

*Department of Economics, Iowa State University. Email: hyunsuko@iastate.edu.

1 Introduction

Temperature shapes the conditions of work and leisure and is a non-negligible part of local amenities. In spatial equilibrium models, amenities as well as productivity determine where people live, what they earn, and what they pay for housing (Rosen, 1979; Roback, 1982), and climate amenities help explain the redistribution of the U.S. population (Rappaport, 2007). Recent evidence from U.S. counties shows that warmer temperatures reduce net migration, with larger responses to more persistent temperature shifts (Baylis et al., 2025). These studies characterize climate by the level or distribution of temperature, which does not depend on the order in which daily temperatures occur. Yet two places with the same distribution of daily temperatures can differ in how those temperatures unfold over time. A stretch of similar days may ask little of residents, whereas a sequence that alternates sharply may require repeated adjustments in work schedules, heating and cooling, and daily activities. Consecutive-day anomaly changes are also a distinct dimension of climate change. Extreme day-to-day temperature changes have become more frequent and intense in mid- to low latitudes under global warming, and they are associated with higher mortality in China and the United States (Liu et al., 2026). If residents value stable daily conditions, amenity measures based only on the level or distribution of temperature would miss part of how climate change alters the attractiveness of places.

I ask whether such day-to-day changes are associated with subsequent county migration in the United States, conditional on the level, dispersion, and seasonality of temperature. I measure day-to-day temperature swings (denoted J) as the annual mean absolute change between consecutive daily temperature anomalies, which is distinguished from the dispersion of temperature (denoted V), or the annual average of monthly anomaly standard deviations. The two measures are designed to capture distinct features of the temperature variability, since reordering the days within a month leaves its standard deviation unchanged but can alter the day-to-day temperature swings. This distinction lets me examine whether the ordering of realized temperatures carries information about

migration beyond the temperature distribution itself. The expected sign is ambiguous, however, since a change that ends a heat wave or cold spell may make a place more attractive rather than less. Section 2 develops this argument in a simple framework of adjustment costs.

This paper contributes to two literatures. First, it adds a new dimension of temperature to research on climate and migration. Existing studies of U.S. internal migration characterize climate by the level of temperature or by the frequency of extreme days (Baylis et al., 2025; Winkler and Rouleau, 2021), whereas this paper examines whether the sequence of daily temperatures carries additional information. Second, it contributes to research on temperature variability by separating the ordering of daily temperatures from their dispersion. Kotz et al. (2021) show that day-to-day temperature variability, measured as the within-month standard deviation of daily temperatures, reduces economic growth. Although labeled day-to-day variability, this measure captures within-month dispersion and is closely related to the dispersion measure used here, so it is unchanged when the days within a month are reordered. Kamukeni and Naito (2024) incorporate longer-horizon temperature variability into an analysis of internal migration in sub-Saharan Africa. The paper also differs from Fan et al. (2025), who develop a dynamic spatial framework in which households learn about future climate paths and damages, so that uncertainty over these paths shapes migration. By contrast, consecutive-day anomaly changes and within-month dispersion summarize observed temperature histories, not households' forecast uncertainty.

I combine daily station records from NOAA's Global Historical Climatology Network (GHCN-Daily) with IRS county-to-county migration data for eight filing-year pairs from 2011–2012 to 2018–2019. Temperature anomalies are measured relative to a 1981–2007 seasonal reference, and the station network is held fixed over the sample period. Requiring complete temperature and precipitation records throughout 2008–2017 yields a panel of 6,805 observations for 859 counties, or 27.6 percent of contiguous U.S. county units,

so the sample is not nationally representative. I regress migration rates on day-to-day swings averaged over the three calendar years preceding the first filing year of each pair, controlling for within-month dispersion, temperature-bin shares, precipitation, and seasonal amplitude measured as the gap between July and January temperatures. County fixed effects absorb persistent local differences, state-by-filing-year fixed effects absorb common state-level shocks, and standard errors are clustered by state. Although swings and dispersion are highly correlated across counties, their correlation falls from 0.94 to 0.56 after removing these fixed effects, which allows their coefficients to be estimated separately.

Averaged over three years, a 1°C increase in consecutive-day anomaly changes is associated with a 0.39 percentage point increase in the out-migration rate, but the estimate is imprecise ($p = 0.149$). Single-year swings and the most recent lag in a distributed-lag model are positively and significantly associated with out-migration, while the sum of the three annual lags is not statistically different from zero. Estimates for in-migration and net migration are sensitive to a single county whose 2011–2012 inflow record appears to reflect an administrative county-assignment anomaly. Within-month dispersion is negatively associated with in-migration in the main specification and when that county is excluded, although this association varies across control sets. Taken together, the results show that the ordering of daily temperatures can be distinguished empirically from the dispersion of temperature, but they provide limited evidence of a stable migration response to day-to-day swings. Because the estimates are conditional associations, they do not identify the adjustment costs through which swings might affect location choice.

The remainder of the paper is organized as follows. Section 2 outlines the economic motivation and defines consecutive-day anomaly changes and within-month dispersion. Section 3 describes the IRS migration data, the construction of weather measures, and the sample. Section 4 presents the empirical strategy. Section 5 reports the results, including

the separability of the two measures, exposure timing, and sample and specification comparisons. Section 6 discusses the findings and concludes.

2 Economic Motivation and Temperature Measures

2.1 Adjustment demands and location choice

Let the daily thermal cost for household h in county c on date d depend on actual temperature T_{cd} , adaptation capital K_{hc} , and an adaptation state q_{hcd} inherited from recent conditions:

$$C_{hcd} = \ell_h(T_{cd}; K_{hc}) + \kappa_h(K_{hc}) \phi_h(T_{cd} - q_{hcd}). \quad (1)$$

The function $\phi_h(\cdot)$ describes how adjustment costs vary with the gap between current temperature and the household's adaptation state, while $\kappa_h(K_{hc}) \geq 0$ scales their magnitude. Adaptation capital may reduce this sensitivity, although we do not impose or estimate a specific functional relationship.

$$q_{hcd} = (1 - \lambda_h)q_{hc,d-1} + \lambda_h T_{c,d-1}, \quad 0 < \lambda_h \leq 1. \quad (2)$$

The first term in equation (1) captures costs associated with the prevailing temperature; the second represents adjustment relative to recent conditions. If $\lambda_h = 1$ and $\phi_h(x) = |x|$, the latter depends on the absolute raw-temperature change from the previous day. With slower adaptation or nonlinear costs, more of the temperature history matters.

A household compares the expected value of income, housing, amenities and thermal costs across locations, net of moving costs and local ties. If repeated changes raise net costs that local adaptation and prices do not offset, they may increase outflows and discourage inflows. This prediction is conditional. Changes can relieve extreme conditions, households differ in exposure, and residential adjustment can occur within a county.

County migration therefore need not respond to the same direction to an absolute-change index.

Evidence that high temperatures affect working time and leisure allocation provides a motivation for behavioral adjustment (Graff Zivin and Neidell, 2014), but does not necessarily leads to a response to J . The framework is deliberately illustrative, as the data contain no direct measure of adjustment costs, beliefs, or the channels through which temperature influences a moving decision. In addition, the economic example concerns actual temperature, while the empirical J removes normal seasonal progression. Thus J is a proxy for one dimension of adjustment demands, but J alone cannot explain the aggregate temperature-related cost of a household.

2.2 Anomalies, consecutive pairs and dispersion

Let s denote a station and y a weather year. Daily mean temperature is the average of TMAX and TMIN. The anomaly is the difference from the station's fitted seasonal reference $\mu_s(d)$:

$$a_{sd} = T_{sd} - \mu_s(d). \quad (3)$$

Define $\mathcal{D}_{sy}$ as the set of dates d in year y for which both d and $d - 1$ have valid observations at station s . The pair is formed by calendar adjacency, not by taking the previous nonmissing observation. Then

$$J_{sy} = \frac{1}{|\mathcal{D}_{sy}|} \sum_{d \in \mathcal{D}_{sy}} |a_{sd} - a_{s,d-1}|. \quad (4)$$

Valid pairs crossing a month or year boundary are retained and assigned to the year of the second date. Missing days are never bridged.

For dispersion, let n_{sym} be the valid-day count in month m and $\bar{a}_{sym}$ its mean anomaly.

I use

$$V_{sy} = \frac{1}{12} \sum_{m=1}^{12} \left[\frac{1}{n_{sym} - 1} \sum_{d \in (y,m)} (a_{sd} - \bar{a}_{sym})^2 \right]^{1/2}. \quad (5)$$

The twelve monthly sample standard deviations receive equal weight. County annual J and V average these station-level measures over a fixed set of within-county stations, with equal station weights.

For illustration, the four-day anomaly sequences $(-1, -1, 1, 1)$ and $(-1, 1, -1, 1)$ have identical means and dispersion. Their mean absolute changes across the three internal pairs are $2/3^\circ\text{C}$ and 2°C . The example isolates sensitivity to order; the regressions do not experimentally rearrange a fixed temperature distribution. J also does not distinguish warming from cooling or identify forecast errors. Conditioning on V separates the fitted association with consecutive-day changes from measured within-month dispersion, while other unmeasured features of the temperature path may remain relevant.

3 Data and Sample

3.1 Migration data

IRS county migration records compare addresses on returns linked across successive filing years. I use eight pairs, from 2011–2012 through 2018–2019, beginning with the IRS’s expanded matching methodology. These labels refer to filings and reported addresses, not a uniform interval during which all individual moves occurred (Internal Revenue Service, n.d.a).

Let I_{ct} , O_{ct} and S_{ct} be the published domestic inflow, domestic outflow and nonmigrant N2 counts for a pair whose first filing year is t . N2 is the IRS count of exemptions or represented individuals, used here as a person-count proxy. It is not a Census population count. In particular, N2 reflects individuals on the second-year returns of matched tax

units, so the denominator is an origin-based matched-record base rather than the county's exact population immediately before migration. I define

$$\begin{aligned}
B_{ct} &= O_{ct} + S_{ct}, \\
M_{ct}^{\text{out}} &= 100 O_{ct} / B_{ct}, \\
M_{ct}^{\text{in}} &= 100 I_{ct} / B_{ct}, \\
M_{ct}^{\text{net}} &= 100 (I_{ct} - O_{ct}) / B_{ct}.
\end{aligned} \tag{6}$$

Positive net migration denotes net inflow. I use official domestic totals rather than sums of published bilateral flows, which are subject to disclosure restrictions. Domestic totals include exchanges with Alaska and Hawaii even though focal counties are in the contiguous United States. Foreign outflows are excluded from B .

The inflow outcome measures intensity relative to the origin-based denominator; it is not an individual's probability of entering the county, and it can exceed 100%. N1 comparisons use the number of returns and a separately constructed N1 denominator. They measure return-based mobility, not the same outcome with a different label.

For any annual weather variable Z , the main exposure is

$$Z_{ct}^{(3)} = \frac{1}{3} \sum_{\ell=1}^3 Z_{c,t-\ell}. \tag{7}$$

Thus the 2011–2012 IRS pair is linked to 2008–2010 weather, and 2018–2019 to 2015–2017. The one-year comparison uses $t - 1$. This alignment places weather before the first filing year, but heterogeneous filing dates and address updates prevent a guarantee that it precedes every migration decision.

3.2 Weather construction

Temperature comes from NOAA GHCN-Daily (Menne et al., 2012; National Centers for Environmental Information, n.d.). I retain daily TMAX and TMIN with blank quality flags and TMAX no lower than TMIN. A fixed contemporary network satisfies annual and monthly completeness requirements throughout 2008–2017. The seasonal reference uses 1981–2007, which does not overlap these exposure years. At least twenty historical years must support each non-leap calendar date. I fit an intercept and four sine–cosine harmonic pairs to the calendar-date means, weighting by historical support.

The temperature controls are daily shares in ten raw-temperature bins; the 15–20°C bin is omitted. They allow different associations across parts of the distribution without assuming that equal changes in annual mean temperature imply equal exposures. Mean temperature is not separately included in the main model, but comparisons add it to the bins or substitute it for them. Bins do not fully control temperature variation within each interval.

Seasonality, or annual temperature spread, captures another time scale. Let $\bar{T}_{cym}$ be county mean temperature in month m , obtained by averaging station monthly means. The main amplitude and its alternative are

$$\begin{aligned} A_{cy}^{JJ} &= \bar{T}_{cy,7} - \bar{T}_{cy,1}, \\ A_{cy}^R &= \max_m \bar{T}_{cym} - \min_m \bar{T}_{cym}. \end{aligned} \tag{8}$$

The main definition is a signed July-minus-January difference; all observed values are positive. These realized annual amplitudes are averaged over the same exposure years as J and V . A fixed historical amplitude would be absorbed by county effects. Realized amplitude adds information about the seasonal arrangement of temperatures, but does not capture every aspect of seasonality.

Precipitation is measured using a separate fixed network within each county. The main controls are mean daily rainfall P and the mean of squared daily rainfall Q . The latter retains information about large daily amounts and is distinct from the square of mean rainfall. All temperature and precipitation controls use the same exposure window as J .

3.3 Sample construction and geographic coverage

Table 1 separates losses in geographic coverage from missing county-period observations. The contemporary temperature network initially covers 1,007 county units. Seasonal-reference support retains 1,718 stations in 1,005 counties. Adding the complete precipitation exposures leaves 859 counties. These counties provide 6,872 potential observations over eight filing-year pairs, of which 67 lack usable IRS outcomes or geographic links. The final sample has 6,805 observations in 48 states: 842 counties have all eight pairs and 17 have fewer.

Table 1. Sample construction

Stage	Counties	County-periods
2019 contiguous-U.S. geographic universe	3,108	—
Fixed contemporary temperature network	1,007	—
Supported 1981–2007 seasonal reference	1,005	—
Complete temperature and precipitation exposures	859	6,872
Usable IRS outcomes and geographic links	859	6,805
Balanced-panel comparison only	842	6,736

Notes: The geographic universe includes county equivalents and the District of Columbia; the final sample spans 48 states and excludes DC. The fixed temperature network has 1,745 stations before seasonal-reference support and 1,718 afterward. The weather-complete set supplies eight potential exposure windows per county. Missing or unusable IRS records remove 67 county-periods, but no additional whole county. The balanced sample is a comparison, not another restriction on the main sample.

Station locations and maps use 2019 Census county boundaries (U.S. Census Bureau, 2019). Code changes are linked where geography is unchanged, while unresolved boundary changes are excluded. The final sample represents 27.6% of the 3,108 contiguous-U.S. county units in the geographic universe. This is county coverage, not population coverage.

Complete weather networks are unevenly distributed, and the resulting sample is not a random national sample. The estimates therefore describe included county-periods rather than a population-weighted national migration response.

Table 2 reports descriptive statistics. Mean three-year J is 2.476°C and mean V is 3.799°C . Mean outflow intensity is 6.003%, while the inflow and net distributions contain a large upper-tail observation. Appendix Figure 1 shows the final sample's geographic coverage.

Table 2. Descriptive statistics

Variable	Mean	SD	Minimum	Maximum
Outflow rate (percent)	6.003	2.047	1.491	27.499
Inflow intensity (percent)	6.164	6.179	1.320	478.566
Net migration rate (percent)	0.162	5.841	-8.058	470.080
$J^{(3)}$ ($^{\circ}\text{C}$)	2.476	0.456	0.931	3.714
$V^{(3)}$ ($^{\circ}\text{C}$)	3.799	0.621	1.503	5.385
Mean temperature ($^{\circ}\text{C}$)	11.361	5.171	-0.366	26.315
July–January amplitude ($^{\circ}\text{C}$)	23.736	5.365	3.726	37.156
Mean daily precipitation (mm)	2.598	1.108	0.128	11.312

Notes: Statistics weight each of the 6,805 county-period observations equally. Weather statistics use three-year exposures. Mean temperature is descriptive and enters only selected comparisons. Inflow and net maxima reflect the Cloud County observation; their common denominator is domestic outflow plus nonmigrants, not destination population.

4 Empirical Strategy

4.1 Main specification

For each outcome $k \in \{\text{out}, \text{in}, \text{net}\}$, I estimate unweighted fixed-effects OLS:

$$\begin{aligned}
M_{ct}^k = & \beta_J^k J_{ct}^{(3)} + \beta_V^k V_{ct}^{(3)} \\
& + \eta^k A_{ct}^{JJ,(3)} + \theta^{k'} H_{ct}^{(3)} + \pi_1^k P_{ct}^{(3)} + \pi_2^k Q_{ct}^{(3)} \\
& + \alpha_c^k + \delta_{\text{state}(c),t}^k + \varepsilon_{ct}^k
\end{aligned} \tag{9}$$

where H contains nine temperature-bin shares. $P_{ct}^{(3)}$ and $Q_{ct}^{(3)}$ are the three-year averages of annual mean daily precipitation and annual mean squared daily precipitation. County fixed effects, denoted as α , absorb persistent local characteristics including fixed climate attributes. State-by-first-filing-year fixed effects, denoted as δ , absorb shocks common to counties in the same state and year. Identification of the linear conditional association uses within-county weather changes relative to those shared patterns. All regressions use state-clustered standard errors.

Because all three N2 regressions use identical observations, weights and regressors, equation (6) implies that each net coefficient equals its inflow coefficient minus its outflow coefficient. Net standard errors are nevertheless estimated directly, accounting for the covariance between inflow and outflow. Net results describe the balance of mobility rather than an independent source of evidence.

4.2 Grouped comparisons and exposure timing

While the three-year average is the primary reporting benchmark, comparisons are grouped by the question they address. Temperature-control comparisons omit J or V , vary the treatment of mean temperature and bins, and omit or redefine amplitude. Timing comparisons use one-year exposure or three separate annual lags. Sample comparisons retain the balanced panel or exclude Cloud County. Additional comparisons replace state-year effects with national-year effects, change the rainfall second moment to squared mean rainfall, and use N1 outcomes. These are related comparisons in an exploratory study, not independent replications.

For the timing comparison, let $\mathbf{Z}_{cy}$ collect all the annual weather regressors in the main specification. The distributed-lag model is

$$M_{ct}^k = \alpha_c^k + \delta_{\text{state}(c),t}^k + \sum_{\ell=1}^3 \mathbf{z}_{c,t-\ell}' \mathbf{b}_\ell^k + u_{ct}^k. \quad (10)$$

All weather variables receive three separate lags, giving 42 weather slopes in total. I report tests of equality for J and V separately and for the entire weather vector. The sum of the three J coefficients describes a one-degree increase in each of the three exposure years, not the long-run effect of a permanent shock. Timing models use the same 6,805 observations, so their differences do not arise from changing the sample.

5 Results

5.1 Geography, correlation and estimating variation

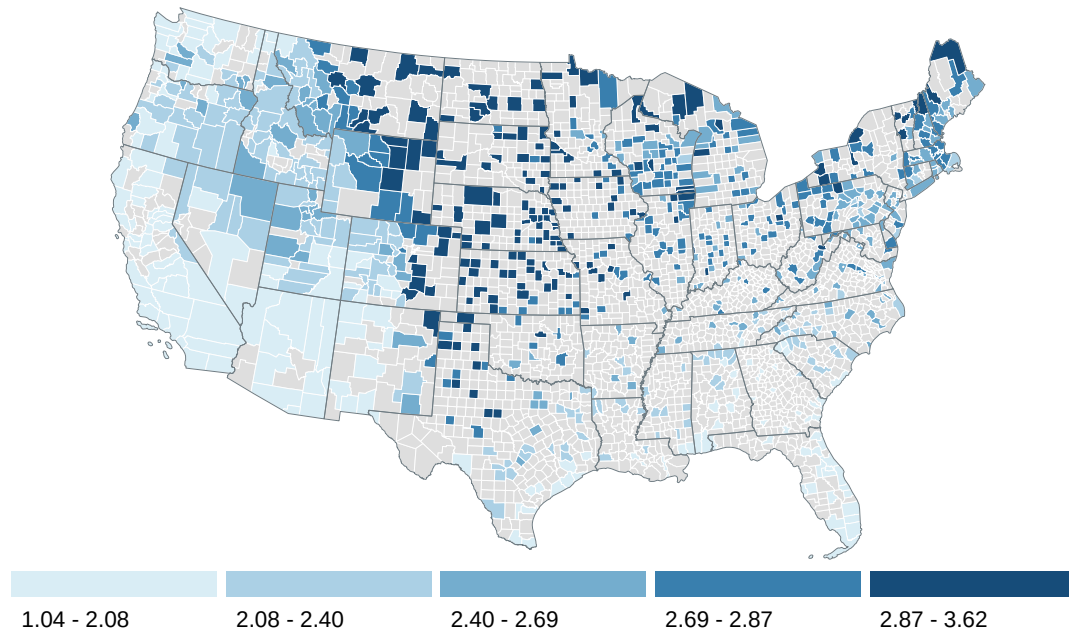
Figure 1 maps each sample county's mean annual J and V over 2008–2017. Each year receives equal weight. This avoids allowing overlapping three-year exposure windows or missing IRS periods to change the descriptive county weights. Higher values are common in parts of the interior and northern United States, while many coastal and southern counties have lower values. The maps describe included counties; grey areas are outside the analysis sample.

The county-mean correlation between J and V is 0.944. In the joint-tercile map (Appendix Figure 2), 690 of 859 counties, or 80.3%, occupy the same tercile for both measures. The other counties occupy adjacent terciles. This substantial geographic overlap is consistent with a shared broad climate pattern, while differences in rank show that the measures are not identical.

The relevant collinearity question, however, concerns the variation used by the regression. The pooled three-year exposure correlation is 0.935; after removing county and state-year effects it is 0.562. Figure 2 displays this distinction. The county-mean correlation of 0.944 and the pooled value of 0.935 differ because they average and weight observations differently.

Table 3 reports variance inflation factors (VIFs) after fixed-effect removal. In the main model, J and V VIFs are 1.49 and 1.57, the maximum VIF is 3.10, and the fourteen weather

A. J: consecutive-day anomaly changes



B. V: within-month anomaly dispersion

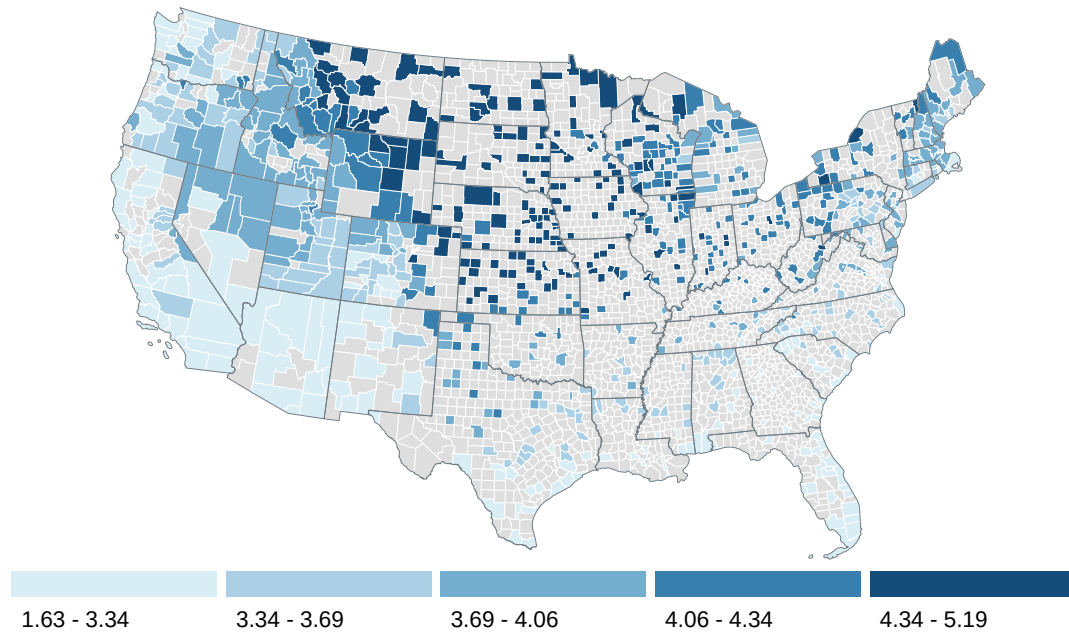


Figure 1. Geographic distribution of annual J and V

Notes: Each map uses 2008–2017 county means and its own quintiles, with thresholds in °C. Equal colors indicate ranks within a measure, not identical temperatures. Grey counties are outside the 859-county sample. Source: GHCN-Daily and 2019 Census boundaries. These are between-county patterns, not estimated migration effects.

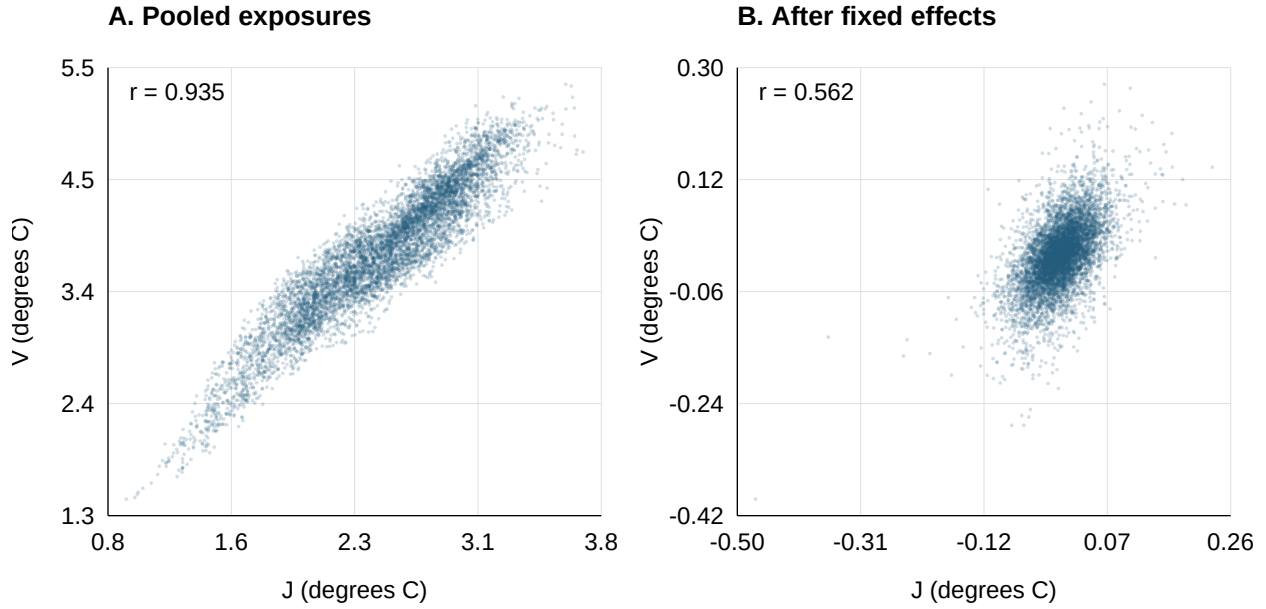


Figure 2. J and V before and after fixed-effect removal

Notes: Both panels use the same 6,805 county-period observations. The right panel removes county and state-by-first-filing-year effects from each three-year exposure. It does not remove the other weather controls. Axis scales differ across panels.

columns have full rank. Adding mean temperature to the bins raises the maximum VIF to 18.33, while the J and V VIFs remain close to their main values. The strongest redundant information is therefore between mean temperature and the distribution controls, rather than a near-linear dependence that prevents separate estimation of J and V.

This does not mean that much variation remains. The standard deviation of three-year J falls from 0.45556°C in levels to 0.03781°C after fixed effects and 0.03098°C after all other regressors. The last quantity is about 81.9% of the fixed-effect residual standard deviation, but only a small fraction of the level variation. High geographic similarity does not invalidate the regression; modest residual variation does limit precision and the scale over which a slope is empirically supported.

Table 3. Collinearity and the scale of estimating variation

Measure	$J^{(3)}$	$V^{(3)}$	
Level SD	0.45556	0.62093	
SD after fixed effects	0.03781	0.05381	
SD after all other regressors	0.03098	0.04298	
Correlation $J^{(3)}$ and $V^{(3)}$	Levels 0.935	After fixed effects 0.562	
Specification	J VIF	V VIF	Maximum VIF
Main	1.49	1.57	3.10
Main plus mean temperature	1.49	1.57	18.33
Mean temperature replaces bins	1.47	1.47	2.43

Notes: All diagnostics use the main sample. Fixed-effect residuals remove county and state-by-first-filing-year effects. Partial SD also removes all remaining weather regressors. VIF is calculated from the fixed-effect residuals, conditioning on the other regressors. The main 14-column weather design has full rank. These diagnostics concern numerical separation and precision, not causal exogeneity.

5.2 Main outflow, inflow and net estimates

Table 4 presents the main results. The J outflow coefficient is 0.390 percentage points per 1°C, with a 95% interval of $[-0.145, 0.926]$. The sign is consistent with higher adjustment demands encouraging departures, but the estimate does not determine the sign or magnitude precisely enough to establish that response.

A one-degree change is approximately 32 residual standard deviations in this regression. Scaling the slope to the observed residual standard deviation of 0.03098°C gives an outflow association of 0.0121 percentage points, with an interval of $[-0.0045, 0.0287]$. This is a change in percentage points, not a 0.0121 percent proportional increase. Residual scaling changes the unit used to describe the result; it does not change the fitted regression or its statistical significance.

The J inflow coefficient is -4.011 , and the net coefficient is -4.401 . Both are imprecise and sensitive to the Cloud observation discussed below. They should not be interpreted as evidence of a general loss of destination attractiveness or net population.

Table 4. Main results: three-year weather exposure

	(1) Outflow	(2) Inflow	(3) Net
$J^{(3)}$	0.390 (0.266) [-0.145, 0.926]	-4.011 (4.978) [-14.025, 6.004]	-4.401 (4.874) [-14.206, 5.404]
$V^{(3)}$	-0.221 (0.292) [-0.809, 0.368]	-2.444** (1.204) [-4.866, -0.022]	-2.224* (1.154) [-4.544, 0.097]
Temperature bins	Yes	Yes	Yes
Precipitation moments	Yes	Yes	Yes
July–January amplitude	Yes	Yes	Yes
Separate mean temperature	No	No	No
County effects	Yes	Yes	Yes
State $\times$ filing-year effects	Yes	Yes	Yes
Observations	6,805	6,805	6,805
Counties	859	859	859

Notes: State-clustered standard errors are in parentheses; 95% confidence intervals are in brackets. All three outcomes use the same sample and the same origin-based denominator. Coefficients are percentage points per 1°C. Two-sided p-values use a t distribution with 47 degrees of freedom. No significance stars are used.

The V outflow coefficient is -0.221 and it is statistically insignificant. Its inflow coefficient is -2.444 , while the net coefficient is -2.224 . These conditional dispersion estimates belong to the same specification and influence assessment as the J estimates. Their nominal significance does not establish an independent causal variability mechanism.

5.3 Exposure windows and distributed lags

The three-year average remains a parsimonious summary, while the unrestricted model permits a richer timing pattern at the cost of more parameters. Table 5 reports timing comparisons for J . With one-year exposure, the outflow coefficient is 0.249. In the distributed-lag model, the most recent J coefficient is 0.259, followed by 0.153 and -0.080 for the second and third years. . The results imply that the day-to-day temperature swing

Table 5. Exposure timing: coefficients on J

Specification	Outflow	Inflow	Net
Three-year average	0.390 [-0.145, 0.926]	-4.011 [-14.025, 6.004]	-4.401 [-14.206, 5.404]
One-year exposure	0.249** [0.061, 0.437]	-0.568 [-2.689, 1.553]	-0.817 [-2.962, 1.327]
Distributed lag 1	0.259** [0.059, 0.458]	-1.097 [-4.351, 2.157]	-1.356 [-4.579, 1.867]
Distributed lag 2	0.153 [-0.222, 0.528]	-1.095 [-4.198, 2.009]	-1.248 [-4.165, 1.670]
Distributed lag 3	-0.080 [-0.350, 0.189]	-2.215 [-6.678, 2.248]	-2.135 [-6.559, 2.290]
Sum of three lags	0.331 [-0.248, 0.911]	-4.407 [-15.166, 6.352]	-4.738 [-15.232, 5.755]
Observations	6,805	6,805	6,805
Counties	859	859	859

Notes: Coefficients are percentage points per 1°C; brackets contain 95% confidence intervals. Unless stated otherwise, all models use N2 outcomes, county and state-by-first-filing-year fixed effects, and standard errors clustered by 48 states. All weather controls follow the indicated exposure window. The main model includes J, V, nine temperature-bin shares, two precipitation moments, and July–January amplitude. Distributed lags enter separately for every weather variable (42 slopes), so lag 1 is the calendar year before the first IRS filing year. The lag sum is a one-degree increase in each of three years, not a permanent-shock effect. V estimates appear in Table ??.

affect the outflow migration of the next year. However, the impacts are diluted as the anomalies of the multiple years are considered in the model, whether its represented by average or by distributed lags.

5.4 Sample comparisons and Cloud County

Table 6 compares the main sample with the balanced panel and the sample excluding all eight Cloud County observations.¹ Cloud’s 2011–2012 inflow intensity is 478.6%, combining the reported inflow with a denominator of 1,367. The unusually large ratio

¹I provisionally interpret the exceptional 2011–2012 inflow for Cloud County as reflecting an administrative county-assignment anomaly. Of 6,542 reported domestic inflow exemptions, 6,113 are attributed to Washington County, Kansas; this concentration coincides with abrupt nonmigrant-count discontinuities in Cloud and neighboring counties. This interpretation has not been confirmed by the IRS, and I have not identified an official correction. I therefore retain the published values in the main analysis and report

alone is not a definitional impossibility, but the record has substantial influence on the regression.

Table 6. Sample comparisons

Specification	Outflow	Inflow	Net
<i>J coefficients</i>			
Main (859 counties)	0.390 [-0.145, 0.926]	-4.011 [-14.025, 6.004]	-4.401 [-14.206, 5.404]
Balanced (842 counties)	0.363 [-0.232, 0.958]	-3.836 [-13.684, 6.012]	-4.199 [-13.864, 5.466]
Exclude Cloud (858 counties)	0.402 [-0.126, 0.929]	0.671** [0.034, 1.309]	0.270 [-0.330, 0.870]
<i>V coefficients</i>			
Main (859 counties)	-0.221 [-0.809, 0.368]	-2.444** [-4.866, -0.022]	-2.224* [-4.544, 0.097]
Balanced (842 counties)	-0.241 [-0.837, 0.355]	-2.474* [-4.972, 0.025]	-2.233* [-4.639, 0.174]
Exclude Cloud (858 counties)	-0.226 [-0.818, 0.366]	-1.517*** [-2.176, -0.858]	-1.291*** [-2.009, -0.573]

Notes: Coefficients are percentage points per 1°C; brackets contain 95% confidence intervals. Unless stated otherwise, all models use N2 outcomes, county and state-by-first-filing-year fixed effects, and standard errors clustered by 48 states. All weather controls follow the indicated exposure window. The main model includes *J*, *V*, nine temperature-bin shares, two precipitation moments, and July–January amplitude. The main, balanced and Cloud-excluded samples contain 6,805, 6,736 and 6,797 observations, respectively. Cloud exclusion removes all eight observations for FIPS 20029. The balanced comparison retains only counties with all eight usable IRS pairs. Differences between separately estimated coefficients are descriptive; no cross-model difference test is reported.

Excluding Cloud changes the *J* inflow coefficient from -4.011 to 0.671 . The outflow coefficient changes little, from 0.390 to 0.402 , and remains imprecise. The net coefficient becomes 0.270 . The positive inflow estimate in the exclusion sample does not fit a simple interpretation in which greater changes uniformly deter arrivals. The lack of a precise net coefficient also prevents a firm conclusion about the balance of migration.

The *V* inflow and net coefficients remain negative after exclusion, at -1.517 and -1.291 , while the *V* outflow coefficient is -0.226 and imprecise. This difference reinforces the exclusion of Cloud County as an influence diagnostic. The counts are present in both official flow directions; see Internal Revenue Service (n.d.b).

need to report both measures and all three outcomes, rather than select a single significant coefficient as the paper's main finding.

Restricting the sample to 842 counties with all eight IRS pairs yields a J outflow coefficient of 0.363 and inflow and net coefficients close to the main estimates. Balanced-panel restriction addresses changing availability within the observed sample. It does not make that sample nationally representative or remove administrative anomalies.

6 Conclusion

Consecutive-day changes and within-month dispersion summarize different aspects of realized weather. Their geographic distributions overlap substantially, yet the main specification permits their separate estimation. Remaining variation is small, and statistical separability does not establish causal identification or support extrapolation to large temperature changes.

The main three-year result is a positive but imprecise J outflow association. Shorter exposure windows give more positive evidence for recent changes, but the unrestricted three-lag sum remains uncertain. Inflow and net results depend substantially on an anomalous administrative record. Taken together, the estimates do not establish a stable migration response to repeated temperature changes. while the insignificant coefficients do not demonstrate that the true relationship is exactly zero.

Three features limit the economic interpretation. First, eight filing-year pairs and overlapping exposure windows provide a short history, while tax-record addresses do not reveal exact move dates or decision-time exposure. Changes in matching, geocoding and disclosure practices may affect county totals and denominators, as state-year effects need not absorb county-specific discontinuities (Internal Revenue Service, n.d.a). Second, the sample covers counties with sustained weather networks and therefore has difficulties

with representing a national population average. Third, the regressions do not observe adaptation investments, wages, housing costs or intermediate behavioral responses.

The paper distinguishes realized consecutive-day changes from dispersion and makes the roles of timing, seasonality and influential records explicit. Direct evidence on intermediate costs and a longer, consistently measured migration panel would be needed to strengthen the economic mechanism or a causal claim.

References

- Baylis, P., P. Bharadwaj, J. T. Mullins, and N. Obradovich. 2025. "Climate and migration in the United States." *Journal of Public Economics* 249: 105446. <https://doi.org/10.1016/j.jpubeco.2025.105446>
- Fan, J., S. Hong, and F. Parro. 2025. "Learning and Expectations in Dynamic Spatial Economies." Working paper, September version, hosted in the AEA 2026 conference program. <https://www.aeaweb.org/conference/2026/program/paper/4iDQtyKF>
- Graff Zivin, J., and M. Neidell. 2014. "Temperature and the Allocation of Time: Implications for Climate Change." *Journal of Labor Economics* 32(1): 1–26. <https://doi.org/10.1086/671766>
- Kamuikeni, C., and H. Naito. 2024. "The Effect of Climate Change on Internal Migration: Evidence from Micro Census Data of 16 Sub-Saharan African Countries." Tsukuba Economics Working Papers 2024-003. <https://pepp.hass.tsukuba.ac.jp/RePEc/2024-003.pdf>
- Kotz, M., L. Wenz, A. Stechemesser, M. Kalkuhl, and A. Levermann. 2021. "Day-to-day temperature variability reduces economic growth." *Nature Climate Change* 11: 319–325. <https://doi.org/10.1038/s41558-020-00985-5>
- Menne, M. J., I. Durre, R. S. Vose, B. E. Gleason, and T. G. Houston. 2012. "An Overview of the Global Historical Climatology Network-Daily Database." *Journal of Atmospheric and Oceanic Technology* 29(7): 897–910. <https://doi.org/10.1175/JTECH-D-11-00103.1>
- Rappaport, J. 2007. "Moving to Nice Weather." *Regional Science and Urban Economics* 37(3): 375–398. <https://doi.org/10.1016/j.regsciurbeco.2006.11.004>
- Roback, J. 1982. "Wages, Rents, and the Quality of Life." *Journal of Political Economy* 90(6): 1257–1278. <https://doi.org/10.1086/261120>
- Internal Revenue Service. n.d.a. "SOI Tax Stats — Migration Data," with county migration documentation for the filing-year pairs used in this study. Accessed September 2026. <https://www.irs.gov/statistics/soi-tax-stats-migration-data>
- National Centers for Environmental Information. "Global Historical Climatology Network Daily." Data and documentation. Accessed September 15, 2026. <https://www.ncei.noaa.gov/products/land-based-station/global-historical-climatology-network-daily>
- Internal Revenue Service. n.d.b. "U.S. Population Migration Data, 2011–2012," user guide and county inflow/outflow files. <https://www.irs.gov/pub/irs-soi/1112inpublicmigdoc.pdf>. County files: <https://www.irs.gov/pub/irs-soi/countyinflow1112.csv> and <https://www.irs.gov/pub/irs-soi/countyoutflow1112.csv>.
- U.S. Census Bureau. "2019 TIGER/Line Shapefiles: Counties and Equivalent Entities." https://www2.census.gov/geo/tiger/TIGER2019/COUNTY/tl_2019_us_county.zip
- Rosen, S. 1979. "Wage-based indexes of urban quality of life." In *Current Issues in Urban Economics*, 74–104. Johns Hopkins University Press.
- Liu, Q., C. Fu, Z. Xu, and A. Ding. 2026. "Global warming intensifies extreme day-to-day temperature changes in mid–low latitudes." *Nature Climate Change* 16(1): 69–76.
- Winkler, R. L., and M. D. Rouleau. 2021. "Amenities or disamenities? Estimating the impacts of extreme heat and wildfire on domestic US migration." *Population and Environment* 42(4): 622–648.

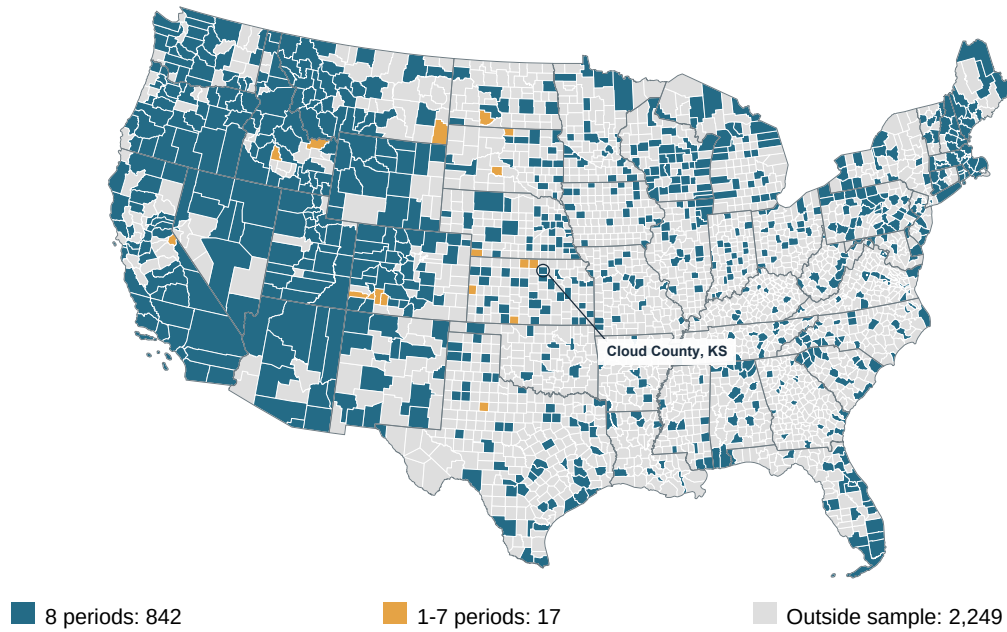


Figure 1. Geographic coverage of the main sample

Notes: Both colored groups belong to the 859-county main sample. All have ten complete annual weather measurements for 2008–2017; the color distinction concerns usable IRS pairs. Grey counties may have weather observations but are outside the final sample. Cloud County is included and labeled. The 3,108-unit universe includes contiguous-U.S. county equivalents and DC.

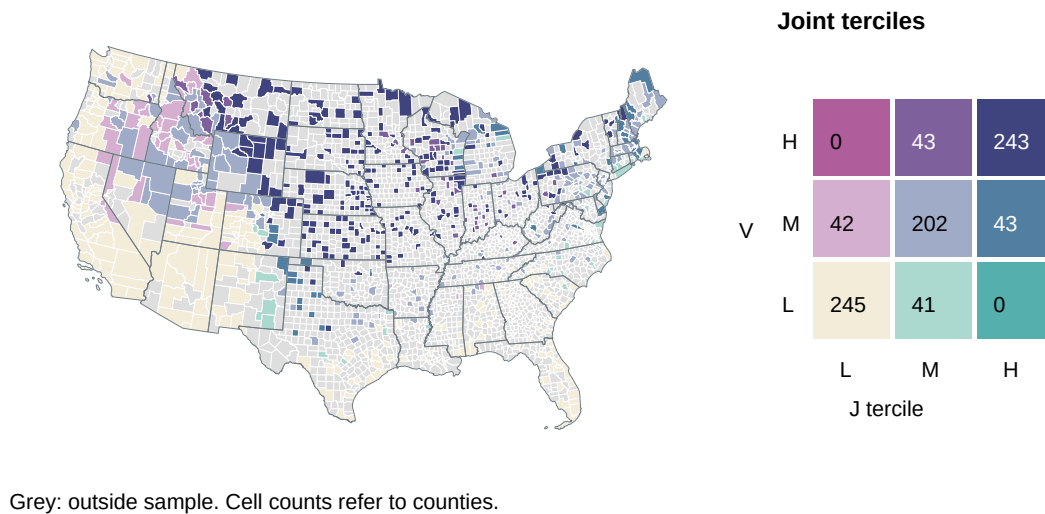


Figure 2. Joint geographic distribution of J and V

Notes: Classifications use separate terciles of county mean annual J and V over 2008–2017. L, M and H denote low, middle and high terciles; numbers inside cells are county counts. The county-mean correlation is 0.944; 690 counties (80.3%) share a tercile and 169 occupy adjacent terciles. No county is in opposite extreme terciles. Small differences around a threshold need not imply large differences in the underlying metrics.